

The hire graph of the 3-free door

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Abstract

Of the two even neighbours of an odd prime $p \neq 3$, exactly one is divisible by 3. The other, the *3-free door* $m_0(p) = p + \chi_3(p)$, is a canonical even integer attached to p whose odd prime factors never include 3. Ownership $q \mid m_0(p)$ places a directed graph on the primes other than 3; adjoining a hub at 2, which divides every door, produces the *hire graph*. On a finite window the graph is the cone on its gold subgraph, so the Laplacian spectrum is the standard cone formula and the algebraic connectivity equals 1 precisely when gold is disconnected. Gold is a DAG: arcs run from larger primes to smaller ones, so the relation is never symmetric. Machine checks find gold disconnected through $X = 10^7$, with satellites on 2-power doors. The first connecting window is the first owner of 46 137 211: $X^* = 92\,274\,421$, with $m_0(X^*) = 2 \cdot 46\,137\,211$ and $m_0(46\,137\,211) = 2^2 \cdot 11 \cdot 1\,048\,573$, so two new gold arcs join the mainland at 11 to the last island $\{524\,287, 1\,048\,573, 8\,388\,593\}$. Gold stays connected through $X = 2.2 \cdot 10^9$. Thus $\lambda_2(H_X) = 1$ fails for some finite X . Every odd prime other than 3 lies in the infinite gold component of 5: a Dirichlet bridge in one of two classes modulo $15q$ yields a common owner of 5 and q . The same Dirichlet counting on the two hire classes modulo $6Q$ gives infinitely many primes hiring any fixed odd Q not divisible by 3. Whether gold is disconnected for infinitely many finite windows remains open.

1 Doors

Let χ_3 be the non-principal Dirichlet character modulo 3: $\chi_3(n) = 0$ if $3 \mid n$, and $\chi_3(n) = (\frac{n}{3})$ otherwise, so $\chi_3(n) = +1$ if $n \equiv 1 \pmod{3}$ and $\chi_3(n) = -1$ if $n \equiv 2 \pmod{3}$. For an odd prime $p \neq 3$ write

$$m_0(p) = p + \chi_3(p), \quad m_1(p) = p - \chi_3(p).$$

The value $m_0(p)$ is the *3-free door* of p ; $m_1(p)$ is the complementary neighbour.

Lemma 1. *$m_0(p)$ is even and not divisible by 3. $m_1(p)$ is divisible by 6. $\{m_0(p), m_1(p)\} = \{p-1, p+1\}$. In particular 3 never divides a 3-free door.*

Proof. The residues of $p-1$, p , $p+1$ modulo 3 are a permutation of $\{0, 1, 2\}$, and $p \not\equiv 0 \pmod{3}$. Hence exactly one neighbour of p is divisible by 3. Both neighbours are even. If $p \equiv 1 \pmod{3}$ then $\chi_3(p) = +1$, so $m_0(p) = p+1 \not\equiv 0 \pmod{3}$ and $m_1(p) = p-1 \equiv 0 \pmod{3}$. If $p \equiv 2 \pmod{3}$ then $\chi_3(p) = -1$, so $m_0(p) = p-1 \not\equiv 0 \pmod{3}$ and $m_1(p) = p+1 \equiv 0 \pmod{3}$. In both cases $m_1(p)$ is even and divisible by 3, hence by 6. $\square$

Examples. $p = 11 \equiv 2 \pmod{3}$, so $\chi_3(11) = -1$, $m_0(11) = 10 = 2 \cdot 5$, $m_1(11) = 12 = 2^2 \cdot 3$. $p = 13 \equiv 1 \pmod{3}$, so $\chi_3(13) = +1$, $m_0(13) = 14 = 2 \cdot 7$, $m_1(13) = 12 = 2^2 \cdot 3$.

Lemma 2. *The four residue classes of odd primes modulo 12 determine the doors as follows.*

$p \pmod{12}$	χ_3	$m_0 \pmod{12}$	$v_2(m_0)$	$m_1 \pmod{12}$	$v_2(m_1)$
1	+1	2	= 1	0	≥ 2
5	-1	4	≥ 2	6	= 1
7	+1	8	≥ 2	6	= 1
11	-1	10	= 1	0	≥ 2

Proof. The classes $\{1, 5, 7, 11\} \pmod{12}$ are the units modulo 12. Each line is the conjunction of Lemma 1 with the unique lifting of that class modulo 4 and modulo 3. Residues 2 and 10 modulo 12 are $2 \pmod{4}$, so $v_2(m_0) = 1$; residues 4 and 8 are $0 \pmod{4}$, so $v_2(m_0) \geq 2$. $\square$

The same residue constraint specialises to two classical families.

Lemma 3. *If p and $p + 2$ are both prime and $p > 3$, then $p \equiv 2 \pmod{3}$ and*

$$m_1(p) = m_1(p + 2) = p + 1.$$

Proof. The pair of residues of $(p, p + 2)$ modulo 3 is either $(1, 0)$ or $(2, 1)$. The first makes $p + 2$ divisible by 3, impossible for $p + 2 > 3$. Thus $p \equiv 2 \pmod{3}$ and $p + 2 \equiv 1 \pmod{3}$, so $\chi_3(p) = -1$ and $\chi_3(p + 2) = +1$. Then $m_1(p) = p + 1$ and $m_1(p + 2) = p + 1$. $\square$

Corollary 1. *If an even integer is not divisible by 3, at most one of its two neighbours is an odd prime greater than 3.*

In other words, the only twin-prime pair not sitting around a multiple of 3 is $(3, 5)$; see Hardy–Wright [3, Ch. 1].

Lemma 4. *If p and $q = 2p + 1$ are both prime and $p > 3$, then $p \equiv 2 \pmod{3}$ and $m_0(q) = 2p$.*

Proof. If $p \equiv 1 \pmod{3}$ then $q = 2p + 1 \equiv 0 \pmod{3}$, impossible for $q > 3$. Thus $p \equiv 2 \pmod{3}$, so $q \equiv 2 \pmod{3}$, $\chi_3(q) = -1$, and $m_0(q) = q - 1 = 2p$. $\square$

Thus a Sophie Germain prime $p > 3$ forces the associated safeprime $q = 2p + 1$ to have 3-free door $2p$. Iterating the same relation produces a Cunningham chain of the first kind.

2 Ownership and the hire graph

An odd prime $p \neq 3$ is an *owner* of an odd prime q when $q \mid m_0(p)$. The resulting directed graph on $\{\text{odd primes}\} \setminus \{3\}$ is the 3-free half of the familiar graph whose edges record prime factors of $p \pm 1$, as in Pratt certificates for $p - 1$ [4] and in the Sophie Germain / Cunningham setting of Lemma 4.

By Dirichlet's theorem [1] on the two admissible classes modulo $3q$,

$$p \equiv 1 \pmod{3}, \quad p \equiv -1 \pmod{q} \quad \text{and} \quad p \equiv 2 \pmod{3}, \quad p \equiv 1 \pmod{q},$$

every odd prime $q \neq 3$ has infinitely many owners. Hence the set of primes that occur as 2 or as an odd prime factor of some $m_0(p)$ is all primes except 3.

Lemma 5. *If $q \mid m_0(p)$ for odd primes $p, q \neq 3$, then*

$$q \leq \frac{m_0(p)}{2} \leq \frac{p+1}{2} < p.$$

In particular the directed ownership graph is acyclic, the relation is never symmetric, and on every finite window the number of gold arcs equals the number of undirected gold edges.

Proof. By Lemma 1, $m_0(p)$ is even and at least 4. An odd prime factor therefore divides $m_0(p)/2$, hence is at most $m_0(p)/2 \leq (p+1)/2 < p$. $\square$

The example $5 \mid m_0(11) = 10$ while $11 \nmid m_0(5) = 4$ is the general case, not a curiosity: ownership always points from a larger prime to a smaller one.

Spectra are taken on finite windows. For $X \geq 5$, let the *owners of the window* be the odd primes $p \neq 3$ with $p \leq X$. Let S_X consist of 2 together with every odd prime factor of $m_0(p)$ as p runs over those owners, and write $n = |S_X|$. Thus S_X is not the set of owners: an owner appears as a vertex only if it arises as an odd prime factor of some door in the window. Since $m_0(p) \leq p + 1 \leq X + 1$, every odd prime in S_X is at most $(X + 1)/2$.

The *hire graph* G_X has vertex set S_X , a spoke from 2 to every other vertex, and a directed *gold arc* $p \rightarrow q$ whenever $p, q \in S_X \setminus \{2\}$ and $q \mid m_0(p)$. Gold is witnessed by doors of owners in the window; the source p is required only to lie in S_X , which already forces $p \leq (X + 1)/2$. The underlying undirected gold graph has an edge $\{p, q\}$ when at least one of the two arcs is present. By Lemma 5 at most one of the two arcs can be present.

The factor 2 divides every door, so it is adjacent to every vertex of $S_X \setminus \{2\}$ independently of gold. That is the reason for the star: finite windows are connected whether or not gold is.

Lemma 6. *There is a gold arc $p \rightarrow q$ if and only if $p \equiv -\chi_3(p) \pmod{q}$.*

Proof. The congruence is $q \mid (p + \chi_3(p))$, i.e. $q \mid m_0(p)$. □

Lemma 7. *Every undirected gold edge $\{p, q\}$ is the base of a triangle 2– p – q in G_X .*

Proof. Both p and q are adjacent to 2 by construction. □

3 The cone spectrum

Let H_X be the combinatorial Laplacian of the undirected graph on S_X (spokes plus undirected gold), and let L'_X be the Laplacian of the undirected gold graph on $S_X \setminus \{2\}$. Thus G_X is the cone on its gold subgraph. We record the block computation of its Laplacian spectrum for completeness; algebraic connectivity is due to Fiedler [2].¹

Lemma 8. $\lambda_{\max}(H_X) = n$. *The vector with value $n - 1$ at 2 and value -1 at every other vertex is an eigenvector for this eigenvalue, independently of the gold edges.*

Proof. Let v be that vector. On the star alone, $H_X v = n v$. Each gold edge has both ends in the set where v is constant, so contributes 0 to $H_X v$. By Lemma 9 the remaining eigenvalues are $1 + \mu_i$, and on a graph with $n - 1$ vertices one has $\mu_i \leq n - 1$, so $1 + \mu_i \leq n$. Hence n is the largest eigenvalue. □

Lemma 9. *Let $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_{n-1}$ be the eigenvalues of L'_X . Then*

$$\text{spec}(H_X) = \{0, n\} \cup \{1 + \mu_2, \dots, 1 + \mu_{n-1}\}.$$

In particular $\lambda_2(H_X) = 1$ if and only if the undirected gold graph on $S_X \setminus \{2\}$ is disconnected.

Proof. Order the vertices as 2 followed by $S_X \setminus \{2\}$. Then

$$H_X = \begin{pmatrix} k & -\mathbf{1}^\top \\ -\mathbf{1} & I + L'_X \end{pmatrix},$$

where $k = n - 1$. Vectors constant on $S_X \setminus \{2\}$ yield eigenvalues 0 and n . On the complement $\mathbf{1}^\top y = 0$ the lower block acts as $1 + \mu$, and the copy of $\ker L'_X$ spanned by $\mathbf{1}$ has already been used. The remaining eigenvalues of L'_X are $\mu_2, \dots, \mu_{n-1}$. The algebraic connectivity of a graph vanishes if and only if the graph is disconnected [2], so $\mu_2 = 0$ if and only if gold is disconnected, and then $\lambda_2(H_X) = 1$. □

¹Door identities and the window cone block are checked in Lean 4 (`Hire/Doors.lean`, `Hire/Lemma8.lean`); the connecting witness at X^* is `Hire/WitnessXstar.lean`; the Dirichlet bridge is `Hire/GoldBridge.lean`. The Mathlib card bound is only the coarse estimate $\lambda \leq |V|$, not the cone identity.

Thus $\lambda_2(H_X) = 1$ is not an independent spectral observation: it is exactly the statement that gold is disconnected on the window.

4 Windows and satellites

Early windows are disconnected, so $\lambda_2(H_X) = 1$ by Lemma 9. Larger machine-checked windows find a first connecting cutoff X^* , after which gold stays connected through the range below. These counts were obtained by an SPF sieve and union-find on the undirected gold graph; they are not Lean-checked.

X	$ S_X $	gold edges	gold CCs	connected?	satellites / note
30	4	1	2	no	{7}
50	6	3	2	no	{7}
100	12	7	4	no	{7, 13, 29}, {17}, {31}
200	18	13	4	no	{7, 13, 29, 53, 97}, {17}, {31}
10^4	362	543	2	no	{257, 4111}
$5 \cdot 10^5$	9292	18335	3	no	{8191, 16381}, {65537}
10^7	125662	278752	3	no	{131071}, {524287}
$X^* - 1$	912341	2161493	2	no	{524287, 1048573, 8388593}
$X^* = 92274421$	912342	2161495	1	yes	first mainland
10^8	980293	2327533	1	yes	
10^9	7877141	19761831	1	yes	
$2.2 \cdot 10^9$	16173663	41258757	1	yes	$2^{31} - 1$ not yet in S_X

Small windows in full. $X = 30$: $S = \{2, 5, 7, 11\}$, gold $\{5, 11\}$. $X = 50$: $S = \{2, 5, 7, 11, 19, 23\}$, gold $\{5, 11\}$, $\{5, 19\}$, $\{11, 23\}$. $X = 100$: $S = \{2, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$, gold $\{5, 11\}$, $\{5, 19\}$, $\{5, 41\}$, $\{7, 13\}$, $\{7, 29\}$, $\{11, 23\}$, $\{19, 37\}$. In every row the number of gold arcs equals the number of undirected gold edges, as required by Lemma 5.

The leftover components are not arbitrary. They sit on primes whose door is a power of 2, together with short owner-chains off those primes.

Lemma 10. *Let $p \neq 3$ be an odd prime. Then $m_0(p)$ is a power of 2 if and only if either*

- $p = 2^k - 1$ is a Mersenne prime (necessarily k an odd prime, $k \geq 3$), in which case $p \equiv 1 \pmod{3}$ and $m_0(p) = p + 1 = 2^k$, or
- $p = 2^k + 1$ is a Fermat prime (necessarily k a power of 2, $k \geq 2$), in which case $p \equiv 2 \pmod{3}$ and $m_0(p) = p - 1 = 2^k$.

In both cases p has gold out-degree zero, so it enters S_X only as a target and is isolated in gold until one of its owners itself lies in S_X .

Proof. The 3-free door is always one of $p \pm 1$. If $p = 2^k - 1$ is prime and $p > 3$, then k is an odd prime, so $2^k \equiv 2 \pmod{3}$ and $p \equiv 1 \pmod{3}$. Thus $\chi_3(p) = +1$ and $m_0(p) = p + 1 = 2^k$. If $p = 2^k + 1$ is prime and $p > 3$, then k is a power of 2 with $k \geq 2$, hence even, so $2^k \equiv 1 \pmod{3}$ and $p \equiv 2 \pmod{3}$. Thus $\chi_3(p) = -1$ and $m_0(p) = p - 1 = 2^k$. Conversely, if $m_0(p) = 2^k$ then either $p + 1 = 2^k$ or $p - 1 = 2^k$.

A prime with $m_0(p)$ a power of 2 has no odd prime factor in its door, so it emits no gold arc. It appears in S_X only when some owner $r \leq X$ has $p \mid m_0(r)$. The arc $r \rightarrow p$ is present in G_X only if that owner r itself lies in S_X . $\square$

The primes of Lemma 10 below 10^6 include $5 = 2^2 + 1$, $7 = 2^3 - 1$, $17 = 2^4 + 1$, $31 = 2^5 - 1$, $127 = 2^7 - 1$, $257 = 2^8 + 1$, $8191 = 2^{13} - 1$, $65537 = 2^{16} + 1$, $131071 = 2^{17} - 1$, and $524287 = 2^{19} - 1$. On the windows above, satellites absorb into the component of 5 and

newer 2-power doors replenish leftovers: $\{257, 4111\}$ at $X = 10^4$; $\{8191, 16381\}$ and $\{65537\}$ at $X = 5 \cdot 10^5$; $\{131071\}$ and $\{524287\}$ at $X = 10^7$. The last island before mainland is $\{524287, 1048573, 8388593\}$ at $X^* - 1$. The cutoff $X^* = 92274421$ is the first owner of 46137211: both are prime, $m_0(X^*) = 2 \cdot 46137211$, and $m_0(46137211) = 2^2 \cdot 11 \cdot 1048573$. Thus the new vertex contributes the gold arcs $46137211 \rightarrow 11$ (into the component of 5) and $46137211 \rightarrow 1048573$ (into the last island), so $|S_X|$ rises by 1, the gold-edge count rises by 2, and the number of components drops from 2 to 1. The island itself is elementary: $524287 = 2^{19} - 1$ has $m_0 = 2^{19}$, while $m_0(1048573) = 2 \cdot 524287$ and $m_0(8388593) = 2^4 \cdot 524287$. Gold stays connected through $X = 2.2 \cdot 10^9$. Let $M_{31} = 2^{31} - 1$. Every owner of M_{31} is at least $2M_{31} - 1 \approx 4.29 \cdot 10^9$, so M_{31} cannot enter S_X before that bound; there is no 2-power door prime between 524287 and M_{31} . Connectedness on $[X^*, 2.2 \cdot 10^9]$ is therefore not threatened by a new Lemma 10 island. The lag throughout is the distinction between owners and S_X : an owner of a 2-power door may exist long before that owner itself lies in S_X .

5 Questions

Lemma 9 equates the spectral and graph forms of the finite-window connectivity question.

Question 1. *Is the undirected gold graph on $S_X \setminus \{2\}$ disconnected for every finite X ? Equivalently, is $\lambda_2(H_X) = 1$ for every finite window?*

Machine checks answer Question 1 in the negative: at $X^* = 92274421$ gold is connected, so $\lambda_2(H_X) \neq 1$. This does not prove that every larger window is connected; it only falsifies the universal claim. In particular, a later 2-power door (once hired into S_X) could open a new satellite.

Question 2. *Does every odd prime other than 3 eventually sit in the infinite component of 5 in undirected gold?*

An ascending owner-chain from q need not meet 5; a common owner of both does. The next theorem supplies one by Dirichlet in two classes modulo $15q$.

Theorem 1. *Let $q \neq 3$ be an odd prime. Then q lies in the same connected component as 5 in the infinite undirected gold graph. Moreover there is a finite X such that 5 and q lie in the same gold component of G_X .*

Proof. The case $q = 5$ is empty. Assume $q \neq 3, 5$. Consider the two residue classes modulo $15q$

$$\begin{aligned} (A) \quad r &\equiv 1 \pmod{3}, & r &\equiv -1 \pmod{5q}, \\ (B) \quad r &\equiv 2 \pmod{3}, & r &\equiv +1 \pmod{5q}. \end{aligned}$$

In both classes $\gcd(r, 15q) = 1$: the first congruence keeps r coprime to 3, and the second keeps r coprime to $5q$. Dirichlet's theorem [1] therefore yields infinitely many primes in each class. Take any such prime r . Then $5q \mid m_0(r)$, so there are gold arcs $r \rightarrow 5$ and $r \rightarrow q$. In the infinite gold graph this is the path $q-r-5$.

For a finite window, let p be an owner of r (again by Dirichlet, as in the ownership section), so $r \mid m_0(p)$. Choose $X \geq \max(p, r)$. Then $p \in S_X$ places r in S_X (not merely $r \leq X$); since $q \mid m_0(r)$, also $q \in S_X$, and both arcs from r are present in G_X . $\square$

Thus Question 2 is answered in the affirmative. The same argument gives the Mersenne and Fermat corollaries: every 2-power-door satellite is absorbed into $\text{comp}(5)$ in some finite window.

The same Dirichlet counting extends from a prime ingredient to any fixed odd factor not divisible by 3.

Lemma 11 (Two-class coverage). *Let $Q > 1$ be odd with $3 \nmid Q$, and write $s = \chi_3(Q) \in \{\pm 1\}$. For every prime $p > 3$,*

$$Q \mid m_0(p) \iff p \equiv 2Q - s \text{ or } 4Q + s \pmod{6Q}.$$

Proof. Write $m_0(p) = tQ$ with t even and $3 \nmid t$ (the door is even and 3-free). Then $t \equiv 2$ or $4 \pmod{6}$, so

$$m_0(p) \equiv 2Q \text{ or } 4Q \pmod{6Q}.$$

In either subcase the residue of $m_0(p)$ modulo 3 is forced by $s = \chi_3(Q)$, and the door rule $m_0(p) = p + \chi_3(p)$ selects the unique neighbour congruent to that residue.

If $m_0(p) \equiv 2Q \pmod{6Q}$, then $m_0(p) \equiv 2s \pmod{3}$. When $s = +1$ one has $m_0(p) \equiv 2 \pmod{3}$, so $\chi_3(p) = +1$ and $p = m_0(p) - 1 \equiv 2Q - s \pmod{6Q}$. When $s = -1$ one has $m_0(p) \equiv 1 \pmod{3}$, so $\chi_3(p) = -1$ and $p = m_0(p) + 1 \equiv 2Q - s \pmod{6Q}$.

If $m_0(p) \equiv 4Q \pmod{6Q}$, then $m_0(p) \equiv s \pmod{3}$ after reducing $4 \equiv 1 \pmod{3}$. When $s = +1$ one has $m_0(p) \equiv 1 \pmod{3}$, so $\chi_3(p) = -1$ and $p = m_0(p) + 1 \equiv 4Q + s \pmod{6Q}$. When $s = -1$ one has $m_0(p) \equiv 2 \pmod{3}$, so $\chi_3(p) = +1$ and $p = m_0(p) - 1 \equiv 4Q + s \pmod{6Q}$.

(The cases are checked in Lean as `two_class_coverage` in `Hire/TwoClassCoverage.lean`.)

□

Corollary 2 (Infinite hire of a fixed ingredient). *Let $Q > 1$ be odd with $3 \nmid Q$. Then infinitely many primes p satisfy $Q \mid m_0(p)$.*

Proof. By Lemma 11, $Q \mid m_0(p)$ if and only if $p \equiv 2Q - s$ or $p \equiv 4Q + s \pmod{6Q}$ with $s = \chi_3(Q)$. Both residues are coprime to the modulus $6Q$. Dirichlet's theorem [1] therefore supplies infinitely many primes in each class. (Lean: `infinite_primes_hire`, `exists_prime_hire_gt` in `Hire/InfiniteHire.lean`.)

□

When Q is an odd prime this recovers the ownership count already stated after the definition of owners; the lemma extends it to composite ingredients such as $Q = 49$.

Question 3. *Is the undirected gold graph disconnected for infinitely many finite X ?*

If there are infinitely many Mersenne primes, Lemma 10 makes each new M_p a sink when it first enters S_X . A satellite interval then exists whenever a $5\text{--}M_p$ bridge is hired only later; that is the expected size of a class modulo $15M_p$, and it holds for every computed M_p , but it is not a formal consequence of the sink lemma alone. Infinitude of Mersenne primes is open, and so is Question 3. The theorem does not give an effective X worth computing: Linnik bounds the first prime in the class modulo $15q$, and that prime still has to be hired.

References

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